CALIFORNIA NORTHSTATE UNIVERSITY COLLEGE of DENTAL MEDICINE

Glass ionomer cements & lining and base materials

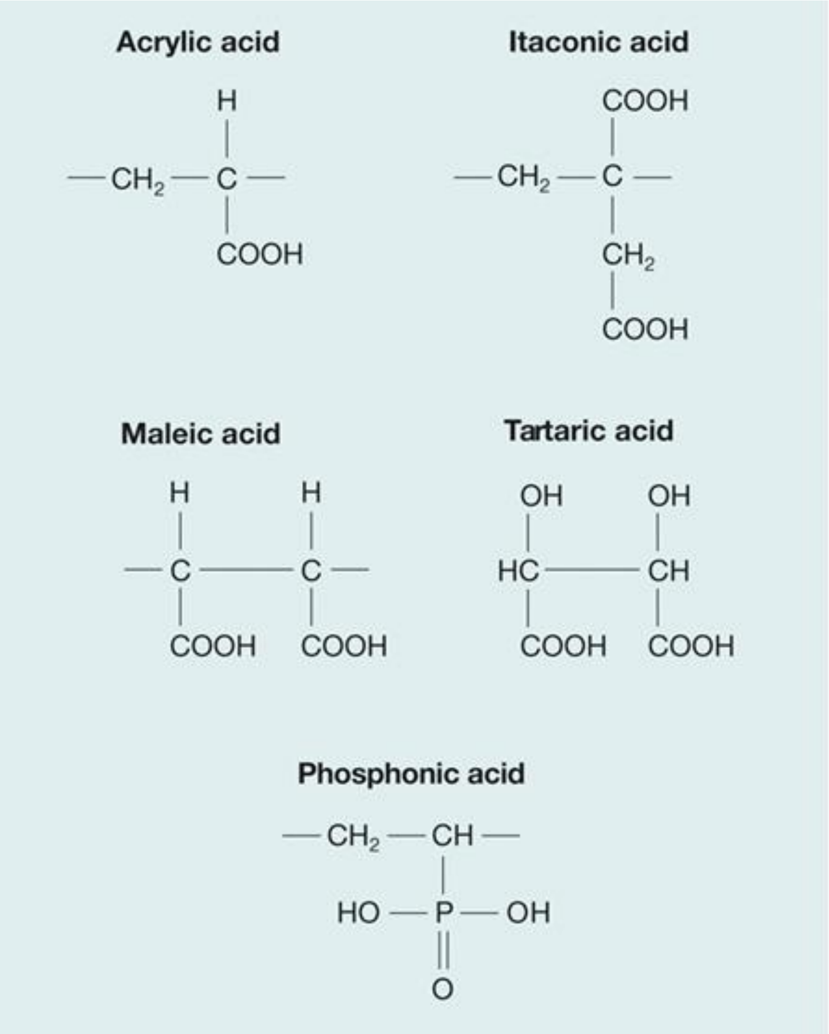
Jayden Yoon, DDS

GIC Intro

- Consists of glass particles that integrate into a network of acidic polymers through acid-base reaction
- Used in most disciplines in dentistry
- Benefits: fluoride releasing, natural adhesion (chelation) to enamel and dentin

GIC components

- Glass
  - Alumina (Al2O3) and silica (SiO2) with fluoride salts
  - Impacts ion release and esthetics
  - Powder form incorporates larger particles for filling GIC vs. smaller for lining GIC
- Polymeric acid (polyacid)
  - Influence setting strength: higher MW --> stronger set
- Water
- Reaction modifiers
  - Tartaric acid most common



GIC forms

- Hand-mixed powder-liquid: strict ratios, mix timing sensitive
- Encapsulated: activation by breaking a membrane separating powder/liquid, then triturating

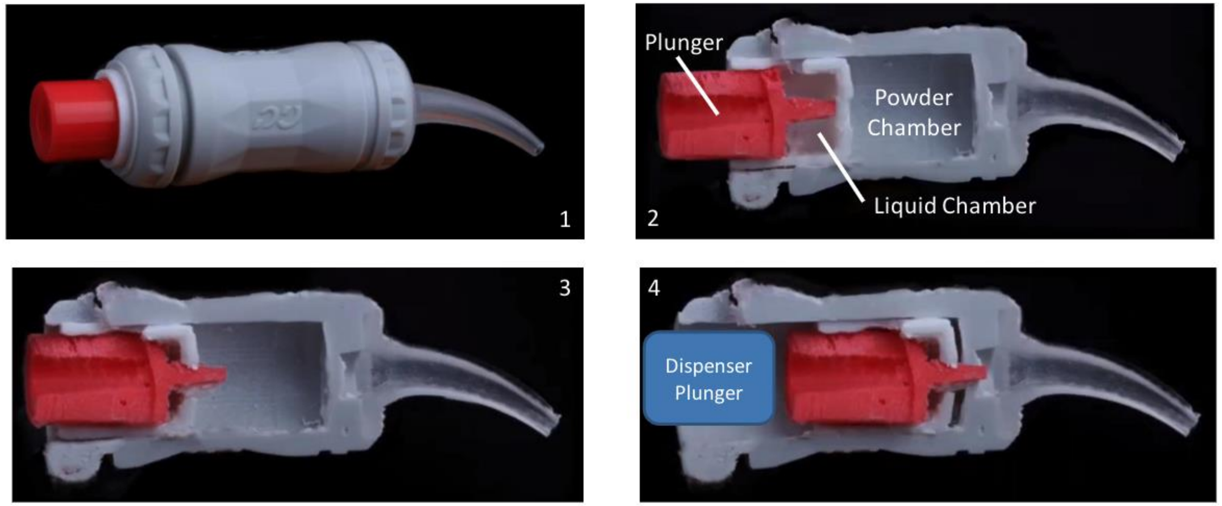


TABLE 2.3.1

Comparison of Properties of Various Formulations and Versions of Glass Ionomers

Property (Unit)	Hand-Mixed GIC	Encapsulated GIC	Hand-Mixed RMGIC	Encapsulated RMGIC
Microhardness (Knoop hardness number) at 1 day	40	58	33	49
Microhardness (Knoop hardness number) at 30 days	47	63	28	37
Biaxial flexural strength (MPa) at 1 day	42	56	123	136
Biaxial flexural strength (MPa) at 30 days	48	52	92	174
Compressive strength (MPa) at 1 day	159	205	165	170
Compressive strength (MPa) at 30 days	156	193	171	182

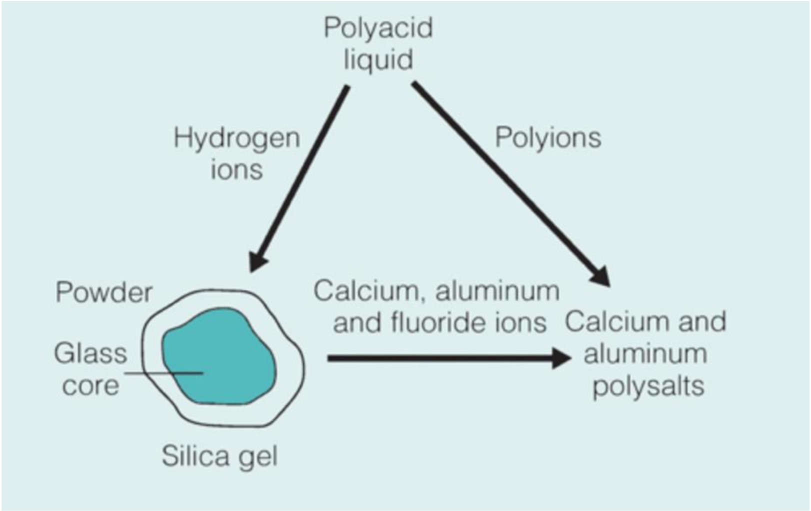
GIC setting reaction

- Acid-base reaction between polymeric acid and the glass
- Calcium and aluminum ions are released from glass to form a salt matrix
- Sodium fluoride is released and flows throughout restoration
- 3 phases
  - Dissolution
  - Gelation
  - Forming

$$\underset{\text{glass}}{MO \cdot SiO_2} + \underset{\text{acid}}{H_2A} \rightarrow \underset{\text{salt}}{MA} + \underset{\text{silica gel}}{SiO_2} + H_2O$$

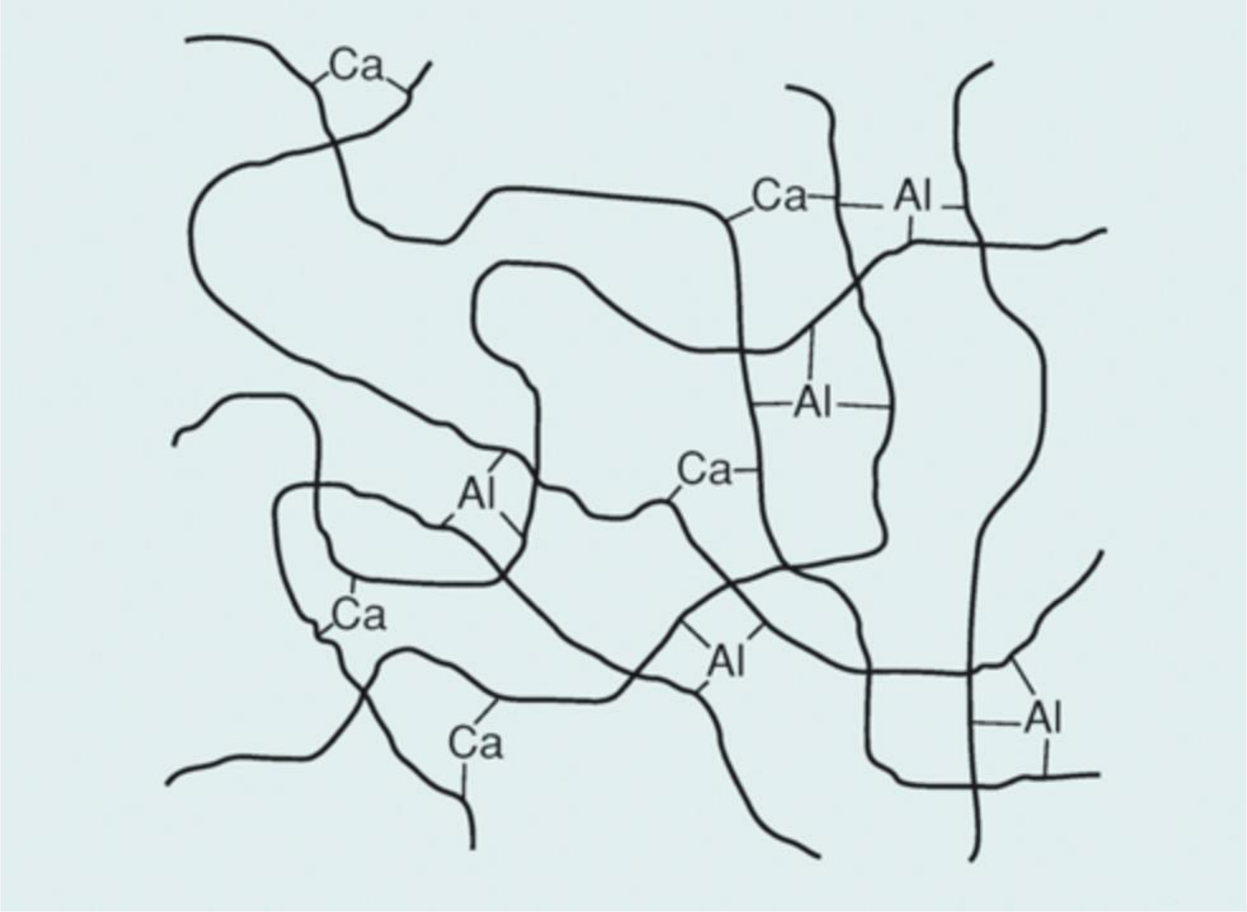
Setting reaction: dissolution

- Acid reacts with outer layer of glass
  - Removes metal ions (aluminum, silica, sodium, fluorine, calcium)
  - Only silica gel remains
  - Hydrogen ions from carboxyl groups diffuse to glass



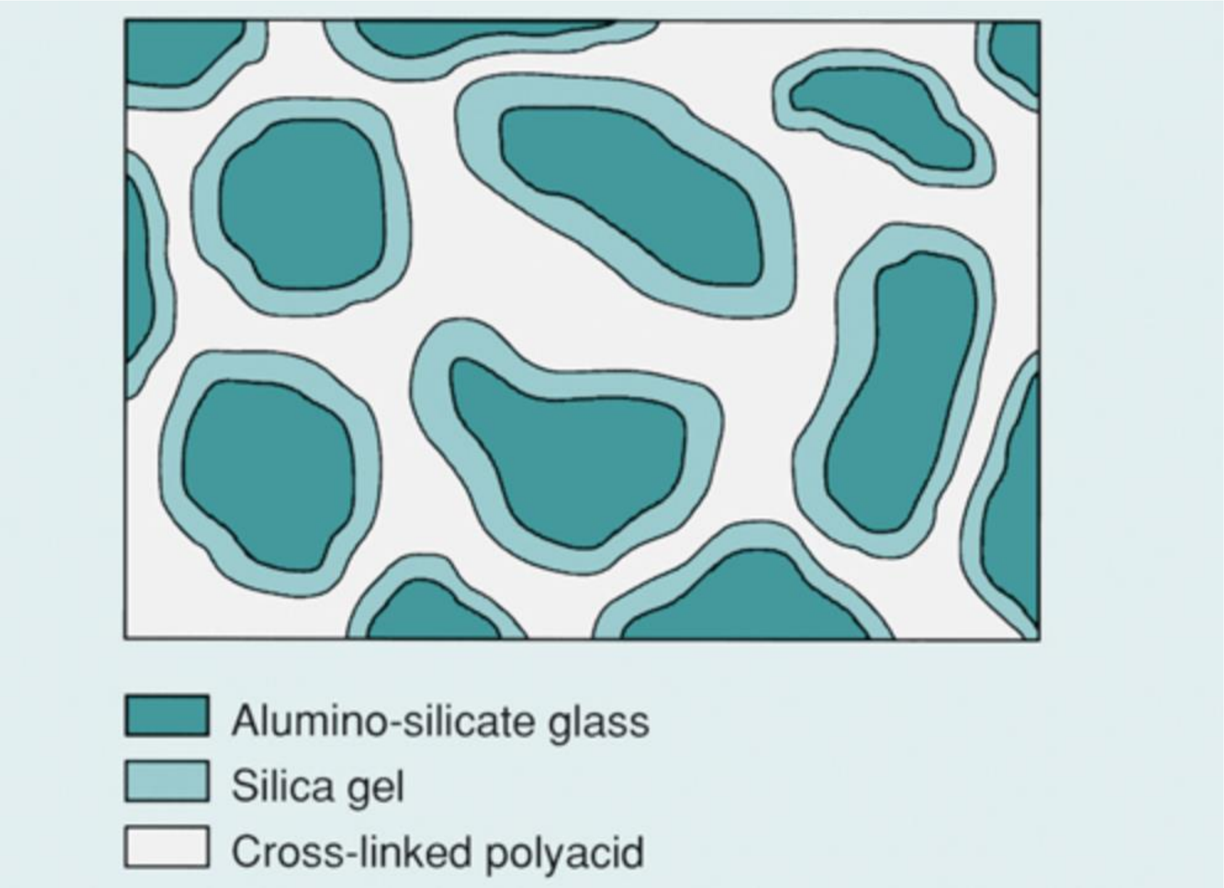
Setting reaction: gelation

- Calcium ions chelate polymeric chains rapidly
  - Aluminum chelates at a slower pace due to trivalency
- Critical notes
  - Needs water to react (do not overly dry restoration)
  - Too much water weakens material and dulls esthetics (reduce water contamination)
  - Aluminum may diffuse out in liquid (cannot form chains)



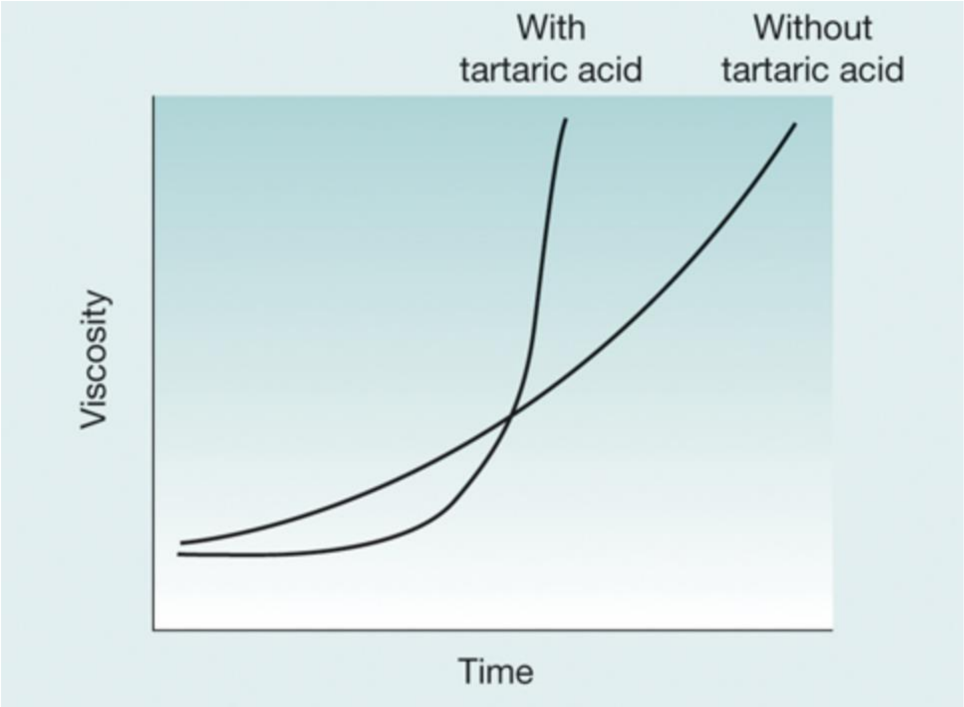
Setting reaction: hardening

- Aluminum leads the hardening reaction with its trivalency (more cross-linking), but is slower than Ca
  - ~24 hrs till completion: hardness, strength, translucency, and esthetics improve
  - Reaction believed to continue for weeks, some dentists add on a layer of varnish to supplement



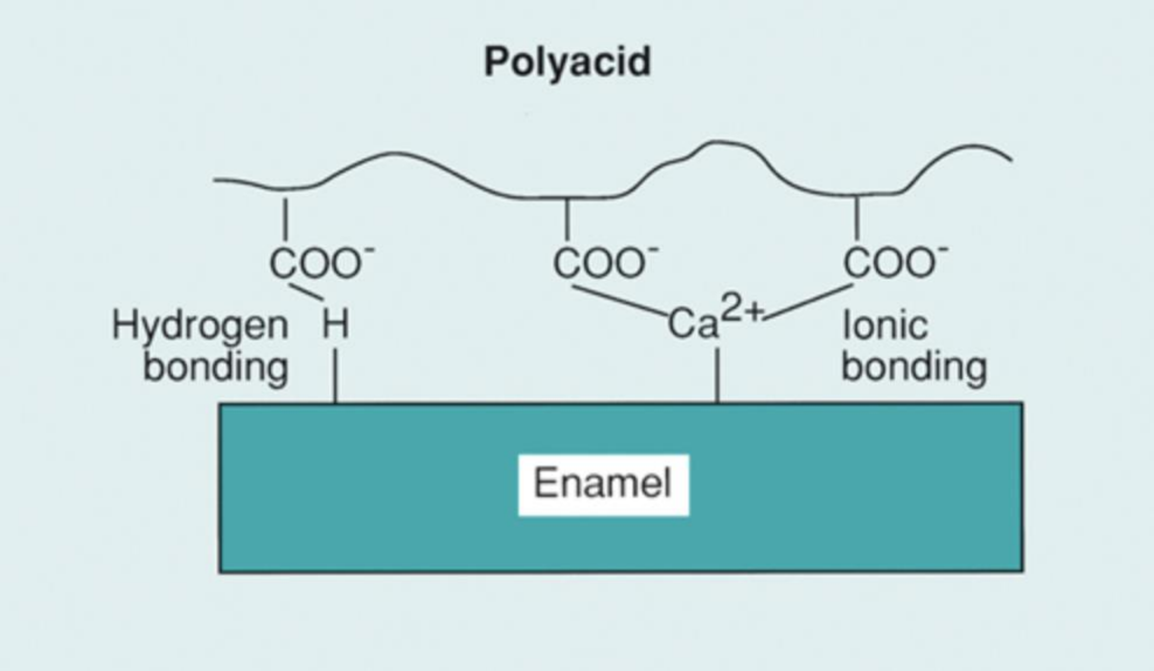
Tartaric acid and water effects

- Tartaric acid
  - Extends working time by slowing down release of Ca ions
  - Then speeds up reaction by increasing rate of reaction of Al cross-linking
- Water
  - Allows ions to migrate, complete reaction, and fluoride movement



Adhesive properties

- Adhesion to tooth structure
  - Acidic nature creates microetching of surface, providing micromechanical bonding and ion release
  - Chelating bond: ionic and hydrogen bonds directly to enamel and dentin
- Sometimes uses dentin (cavity) conditioner
  - ~20% polyacrylic acid, removes dentin smear layer
  - Weaker bond than traditional bonding systems, but sufficient

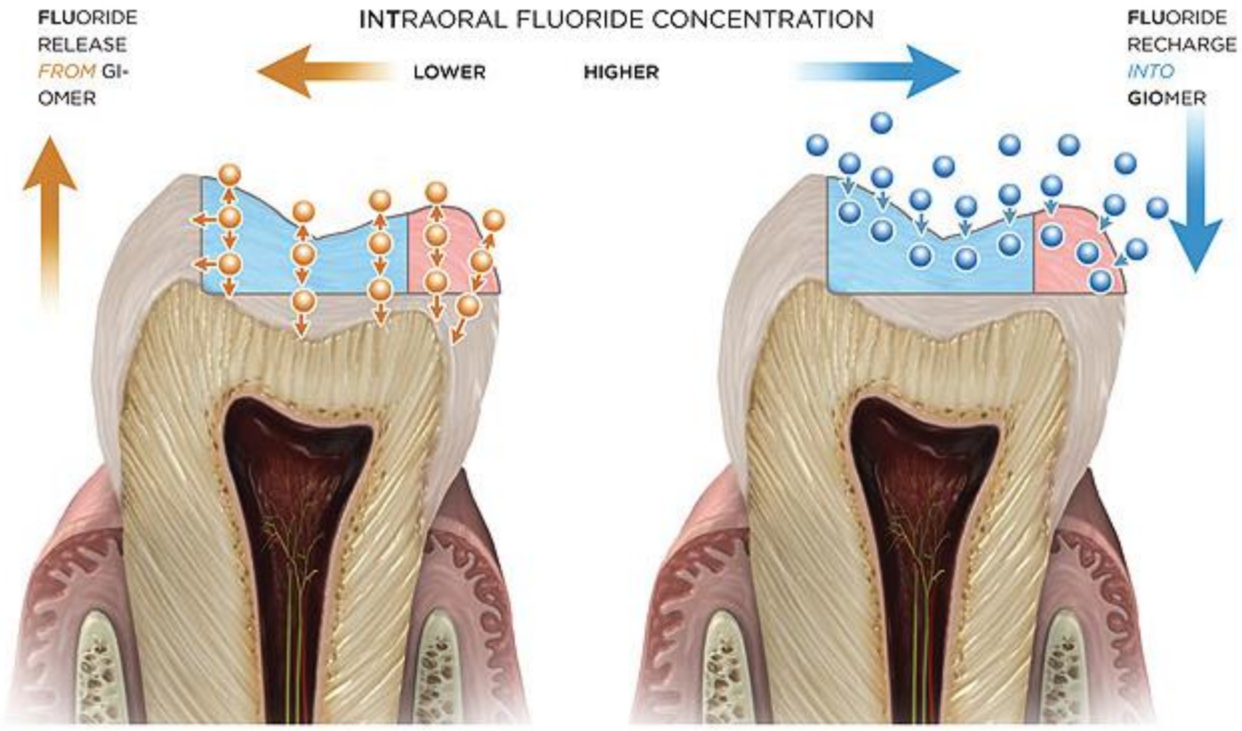


Mechanical properties

- Microhardness increases over time
- Low tensile strength compared to other materials
  - Do not place on high tensile load areas – cusps, incisal edges, etc.

Fluoride release mechanism

- Fluoride within GIC migrates through the water within – can use fluoride from the external environment (saliva)
- Initial release (~4 weeks bolus) is strongest
- Fluoride is concentrated most within the restoration, especially in acidic environments
  - Acid releases more F- because of acid-base reaction in GIC and is concentrated within the restoration
  - Concentration at GIC surface is the least since F- is washed away by saliva



Resin-Modified GICs

- Components
  - Includes water-compatible resin polymers for better handling and mechanical properties
    - Usually hydroxyethyl methacrylate (HEMA) polymer
      - Some safety debates, but no strong evidence against its use
  - Incorporates filler, just like composite, for strength
  - Photoactivator system like composite to assist in handling properties
- Setting reaction
  - Light polymerization occurs first, then the aluminum cross-linking
- Advantages over normal GIC
  - Longer working time with faster set
  - Resin increases strength, resistance to acid

Clinical application of (RM)GICs

- Luting and bonding cements
- Bases or liners in preparations
- Restorations in lower force areas
- Sealants

Part 2 - Cavity lining and base materials

- Liners (<0.5mm thickness) and bases (>0.5mm thickness) are used to protect the pulp from:
  - Thermal stimuli
  - Chemical stimuli
    - Acidity from materials such as zinc-phosphate or zinc-polycarboxylate cements
  - Biological stimuli
    - Microleakage and bacterial entry into crevices
- Material categories
  - CaOH cements
  - Zn-O based cements
  - GI cements
  - RMGICs
  - Visible light-cured resins

CaOH cements (including zinc-oxides)

- Highly basic
- Relatively low compressive strength (~20 MPa), thus mainly used as a liner
- Causes formation of reparative dentin – a calcified layer up to 1.5mm thick
- Consists of 2 pastes:
    - 1: CaOH, zinc oxide, sulfonamide
    - 2: Butylene glycol disilcyate, with other minor chemicals
- Main reaction is a chelating reaction between zinc oxide and butylene glycol disalicylate

Zinc-oxide eugenol cements

- Acid (water) - base (powder) reaction
  - Majority powder, so final product consists of unreacted ZOE held together with salt matrix
- Unmodified ZOE
  - Known for eugenol release - antibacterial properties
  - Poor mechanical strength (compressive strength ~15 MPa) and disintegrates easily – very few uses
- Modified ZOE
  - Additions of hydrogenated rosin to powder and polysterene / methyl methacrylate to liquid
  - Improves mechanical strength and dissolution resistance
  - Can be used as base or liner (compressive strength ~40 MPa)
- EBA cements
  - ZOE with ethoxybenzoic acid (EBA) - improved mechanical properties to be used as base or liner
- Eugenol reacts with free radicals in resins, thus inhibiting polymerization of resin-based materials – be careful of use

$$\underset{\text{base}}{MO} + \underset{\text{acid}}{H_2O} \rightarrow \underset{\text{salt}}{MA} + \underset{\text{water}}{H_2O}$$

(RM)GIC bases and liners

- Naturally, bonds to tooth structure
  - Can be used under amalgams and composites (composites can bond to GICs as well)
    - To bond with composite, etch the GIC alongside the enamel
      - Etching too long will erode the GIC
  - RMGICs do not need an etched surface due to stronger cohesive strength